

Formulation 12_Loading_Rate

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I Objective

The purpose of this document is to extend the model of Garagash (2021) to the case that loading rate is non-zero. It is assumed the reader is familiar with the model of Garagash (2021) as only the necessary aspects related to the extension will be addressed.

2 Formulation

2.1 Equation of Motion

The Equation of Motion from Garagash (2021) reads,

$$K_{\Delta\tau} (\Delta f_0/b, \ell, v_r) + K_{\Delta T} (\ell, \Delta T) - K_c (v_r) = 0, \quad (1)$$

where $K_{\Delta\tau}$ and $K_{\Delta T}$ are the stress intensity factors due to the background stress and impulse, respectively. K_c is the fracture toughness. $\Delta f_0/b$, ℓ , v_r , and ΔT are the overstress, crack length, rupture velocity, and hypocentral impulse size, respectively. In the case of loading rate, the impulse will be dropped (although it can be readily included) and the overstress becomes a function of time, assumed to evolve as $\frac{d(\Delta f_0/b)}{dt}$, such that the Equation of Motion reads,

$$K_{\Delta\tau} (\Delta f_0/b, \ell, v_r) - K_c (v_r) = 0. \quad (2)$$

3 Solving

It is helpful to write the Equation of Motion (EoM) in residual form,

$$r = K_{\Delta\tau} (\Delta f_0/b, \ell, v_r) - K_c (v_r), \quad (3)$$

where r is the residual to be minimized.

3.1 Initialization

To initialize the solver, and initial crack length is proposed and the corresponding rupture velocity is solved for. This is done by finding the rupture velocity that minimizes the residual of Equation 3. Once found, the initial rupture velocity is passed with the initial crack length to the main solver. Note that if the initial crack length is too small, no solution will exist.

3.2 Explicit Solver

Beginning with Equation 3, setting the residual $r (\ell, v_r, \Delta f_0/b) = 0$ and taking the derivative with respect to ℓ yields,

$$\frac{\partial r}{\partial \ell} + \frac{\partial r}{\partial v_r} \frac{\partial v_r}{\partial \ell} + \frac{\partial r}{\partial (\Delta f)} \frac{d(\Delta f)}{dt} \frac{dt}{d\ell} = 0, \quad (4)$$

where $\Delta f = \Delta f_0/b$ for simplicity. Then, by considering,

$$u = \frac{dt}{d\ell} = \frac{1}{v_r}, \quad (5)$$

it follows through differentiation that,

$$\frac{dv}{d\ell} = -\frac{1}{u^2} \frac{du}{d\ell}. \quad (6)$$

By substitution follows,

$$\frac{\partial r}{\partial \ell} - \frac{\partial r}{\partial v_r} \frac{1}{u^2} \frac{du}{d\ell} + \frac{\partial r}{\partial (\Delta f)} \frac{d(\Delta f)}{dt} u = 0, \quad (7)$$

meaning,

$$\frac{du}{d\ell} = \frac{u^2}{\frac{\partial r}{\partial v_r}} \left(\frac{\partial r}{\partial \ell} + \frac{\partial r}{\partial (\Delta f)} \frac{d(\Delta f)}{dt} u \right). \quad (8)$$

Equation 8 is the ordinary differential equation that must be solved by the simulator to find the new rupture velocity corresponding to the next crack length. Note that in the case of no loading rate Equation 8 reduces to the ODE used to solve the formulation of Garagash (2021).

3.3 Implicit Solver

In some instances, in particular for the slip law, the solver becomes very stiff and the solution branch turns back on itself, as $\frac{\partial r}{\partial v_r}$ becomes very small. Ultimately rupture velocity will continue increasing for a decreasing crack length. This phenomenon can be seen in the original formulation of Garagash (2021) in their figure 5a, for example. The explicit solver will not be able to find the solution as written here. Instead, it is desirable to pass to an implicit solver which finds a crack length for a given rupture velocity.

The program should take the last solution correctly solved for by the explicit solver and define a vector of rupture velocities from this crack length up to some maximum crack length, say 90% of the shear wave speed. These rupture velocities are passed to the implicit solver and the corresponding crack lengths are found.

Specifically, when the equilibrium branch becomes multi-valued in crack length, the explicit formulation breaks down because $\partial r / \partial v_r \rightarrow 0$, causing the governing ODE to become singular. The solution is therefore re-parameterized in terms of rupture velocity. Given a prescribed rupture velocity v_i and the converged solution (ℓ_{i-1}, t_{i-1}) from the previous continuation step, the crack length is obtained by solving the nonlinear equation

$$r(\ell_i, v_i, \Delta f_i) = 0, \quad (9)$$

where the overstress is related to the loading history through

$$\Delta f_i = \Delta f_0 + \dot{\Delta f} t_i. \quad (10)$$

Since the travel time depends on the unknown crack length,

$$t_i = t_{i-1} + \frac{\ell_i - \ell_{i-1}}{\bar{v}_i}, \quad \bar{v}_i = \frac{v_i + v_{i-1}}{2}, \quad (11)$$

the quantities ℓ_i , t_i , and Δf_i are coupled. The solution is obtained by fixed-point iteration: an initial estimate of the overstress is used to solve for ℓ_i , the corresponding travel time is updated, a new overstress is computed, and the process is repeated until the change in crack length between successive iterations falls below a prescribed tolerance.

After the last investigated rupture velocity, the program can head back to the explicit solver to finish solving for the last rupture velocities (when $\partial v_r / \partial \ell$ becomes small again). The three solutions (the first explicit, the implicit, and the second explicit solutions) can then be concatenated.

4 Conclusion

The model of Garagash (2021) has been extended to account for a prescribed time-dependent background overstress.

References

- D. Garagash. Fracture mechanics of rate-and-state faults and fluid injection induced slip. *Philosophical Transactions of the Royal Society A*, 379, 2021. doi: <https://doi.org/10.1098/rsta.2020.0129>.